2D Matrices: Diagonalisation

Questions 1–14

Question 1. Matrix $\mathbf{A} = \begin{pmatrix} 4 & 1 \\ 2 & 3 \end{pmatrix}$ has eigenvalues such that $\lambda_1 < \lambda_2$. Matrix $\mathbf{D}$ is diagonal and is defined such that $\mathbf{D} = \mathbf{P}^{-1}\mathbf{AP}$.

a) By finding the eigenvectors of $\mathbf{A}$, find the matrix $\mathbf{P}$.

b) By finding $\mathbf{P}^{-1}\mathbf{AP}$, find the matrix $\mathbf{D}$.

Question 2. For each of the following, find a possible matrix $\mathbf{P}$ and a possible diagonal matrix $\mathbf{D}$ such that $\mathbf{D} = \mathbf{P}^{-1}\mathbf{AP}$:

a) $\mathbf{A} = \begin{pmatrix} 7 & 2 \\ 3 & 6 \end{pmatrix}$

b) $\mathbf{A} = \begin{pmatrix} 20 & 12 \\ 15 & 17 \end{pmatrix}$

c) $\mathbf{A} = \begin{pmatrix} 12 & 7 \\ -14 & -9 \end{pmatrix}$

Question 3. Given the matrix $\mathbf{A} = \begin{pmatrix} 5 & 1 \\ 0 & 5 \end{pmatrix}$

a) Find the eigenvalues and corresponding eigenvectors of the matrix $\mathbf{A}$.
b) Explain why this matrix cannot be diagonalised.

Question 4. Aaron is finding eigenvalues and eigenvectors to diagonalise the matrix $\mathbf{A} = \begin{pmatrix} 13 & 6 \\ 3 & 10 \end{pmatrix}$. His workings are shown below:

$$\lambda = 7,\ \lambda = 16,\ \begin{pmatrix} 1 \\ -1 \end{pmatrix} \text{ and } \begin{pmatrix} 2 \\ 1 \end{pmatrix}$$

$$\mathbf{PDP}^{-1} = \begin{pmatrix} 1 & 2 \\ 1 & -1 \end{pmatrix} \begin{pmatrix} 7 & 0 \\ 0 & 16 \end{pmatrix} \begin{pmatrix} -1 & -1 \\ 1 & 2 \end{pmatrix} = \begin{pmatrix} -30 & 9 \\ 18 & -39 \end{pmatrix}$$

a) Find the two errors in his working.
b) Fully correct his working.

Question 5. Find each of the following:

a) $\begin{pmatrix} 5 & 0 \\ 0 & 4 \end{pmatrix}^8$

b) $\begin{pmatrix} 1 & 0 \\ 0 & -2 \end{pmatrix}^{10}$

c) $\begin{pmatrix} a & 0 \\ 0 & b \end{pmatrix}^k$ for constants a, b and k

Question 6. Show that if $\mathbf{M} = \mathbf{PDP}^{-1}$ then $\mathbf{M}^n = \mathbf{PD}^n\mathbf{P}^{-1}$.

Question 7. Given the matrix $\mathbf{A} = \begin{pmatrix} 3 & 1 \\ 0 & 2 \end{pmatrix}$ and $\mathbf{D} = \mathbf{P}^{-1}\mathbf{AP}$

a) Find the eigenvalues and corresponding eigenvectors of the matrix $\mathbf{A}$.
b) Find possible matrices $\mathbf{P}$ and $\mathbf{D}$.
c) Hence find $\mathbf{A}^4$.

Question 8. Matrix $\mathbf{A} = \begin{pmatrix} -4 & 0 \\ -15 & 1 \end{pmatrix}$ has eigenvalues $\lambda_1 < \lambda_2$.

a) Find the eigenvalues of the matrix $\mathbf{A}$.
b) Find the corresponding eigenvectors for your eigenvalues.
c) Find possible matrices $\mathbf{P}$ and $\mathbf{D}$ such that $\mathbf{P}^{-1}\mathbf{AP} = \mathbf{D}$.
d) Hence find $\mathbf{A}^{50}$, giving each element in terms of suitable powers of 2.

Question 9. Given the matrix $\mathbf{M} = \frac{1}{4}\begin{pmatrix} 3 & 2 \\ 1 & 2 \end{pmatrix}$ has eigenvalues $\lambda_1 = 1$ and $\lambda_2 = \frac{1}{4}$ with corresponding eigenvectors $\mathbf{v}_1 = \begin{pmatrix} 1 \\ 0 \end{pmatrix}$ and $\mathbf{v}_2 = \begin{pmatrix} -1 \\ 2 \end{pmatrix}$.

a) Find $\mathbf{M}^n$ by using $\mathbf{M}^n = \mathbf{P}\mathbf{D}^n\mathbf{P}^{-1}$ where $\mathbf{D}$ is a diagonal matrix and $\mathbf{P}$ is the matrix of eigenvectors.

b) Given that $\mathbf{M}^n \to \mathbf{L}$ as $n \to \infty$, find the matrix $\mathbf{L}$.

Question 10. Matrix $\mathbf{M} = \frac{1}{6}\begin{pmatrix} 4 & 2 \\ 1 & 5 \end{pmatrix}$ has eigenvalues $\lambda_1 < \lambda_2$.

a) Find $\mathbf{M}^n$ by using $\mathbf{M}^n = \mathbf{P}\mathbf{D}^n\mathbf{P}^{-1}$ where $\mathbf{D}$ is a diagonal matrix and $\mathbf{P}$ is the matrix of eigenvectors.

b) Given that $\mathbf{M}^n \to \mathbf{L}$ as $n \to \infty$, find the matrix $\mathbf{L}$.

Question 11. Given the matrix $\mathbf{A} = \begin{pmatrix} 4 & 3 \\ 0 & 2 \end{pmatrix}$. By considering $\mathbf{PD}^n\mathbf{P}^{-1}$ where $\mathbf{D}$ is diagonal and $\mathbf{P}$ is the matrix of eigenvectors, show that

$$\mathbf{A}^n = \begin{pmatrix} 4^n & \frac{3}{2}(4^n - 2^n) \\ 0 & 2^n \end{pmatrix}.$$

Question 12. Given the matrix $\mathbf{A} = \begin{pmatrix} 2 & 6 \\ 1 & 3 \end{pmatrix}$. By considering $\mathbf{PD}^n\mathbf{P}^{-1}$ where $\mathbf{D}$ is diagonal and $\mathbf{P}$ is the matrix of eigenvectors, show that

$$\mathbf{A}^n = 5^{n-1}\begin{pmatrix} 2 & 6 \\ 1 & 3 \end{pmatrix}.$$

Question 13. Given the matrix $\mathbf{A} = \begin{pmatrix} p & q \\ p & q \end{pmatrix}$ where p and q are constants such that $p \neq -q$ and $q \neq 0$, show that

$$\mathbf{A}^n = (p+q)^{n-1}\mathbf{A}.$$

Question 14. Given the matrix $\mathbf{A} = \begin{pmatrix} 1 & k+1 \\ k-1 & -1 \end{pmatrix}$ where k is a constant and $k \neq \pm 1$ and $k \neq 0$, show that

$$\mathbf{A}^{2n+1} = k^{2n}\mathbf{A}.$$